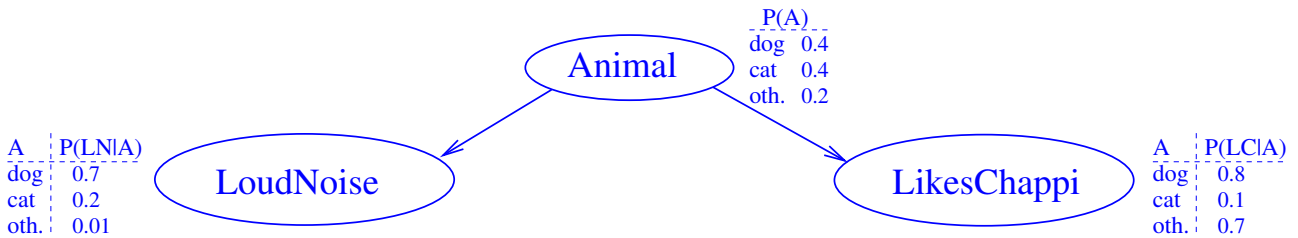


Exercise 1 :

Consider the following Bayesian network BN :



Use inference by enumeration to compute the following probabilities:

- $P(dog \mid loudnoise, likeschappi)$.
- $P(loudnoise \mid \neg likeschappi)$.

Include intermediate steps at a level of granularity as in the lecture slides examples. In particular, for each of (a) and (b), state what the query variable, evidence, and hidden variables are; and write down *which* probabilities provided in BN can be combined *how* to obtain the demanded probability P .

Exercise 2 :

In the graphs in Figures 1 and 2, determine which variables are d-separated from A . Note that it is sufficient to find a single open path along which evidence can be transmitted; if such a path exists then the variables are *not* d-separated (instead they are said to be d-connected).

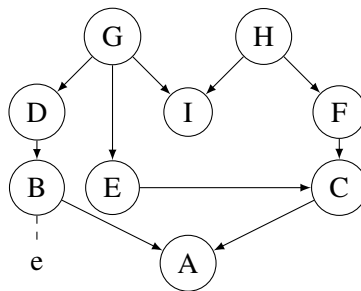


Figure 1: Figure for Exercise 2.

Exercise 3 :

Consider the network in Figure 3.

- What are the minimal set(s) of variables that we should have evidence on in order to d-separate C and E (that is, sets of variables for which no proper subset d-separates C and E)?
- What are the minimal set(s) of variables we should have evidence on in order to d-separate A and B ?
- What are the maximal set(s) of variables that we can have evidence on and still d-separate C and E (that is, sets of variables for which no proper superset d-separates C and E)?

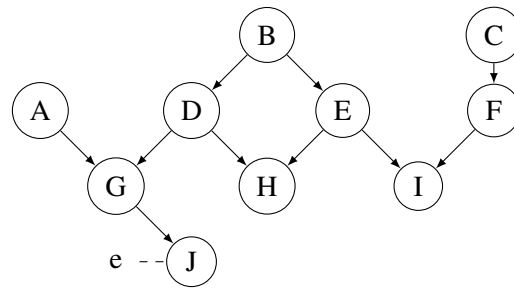


Figure 2: Figure for Exercise 2.

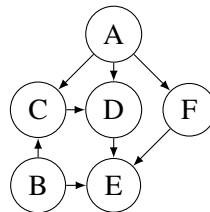


Figure 3: A causal network for Exercise ??.

- What are the maximal set(s) of variables that we can have evidence on and still d-separate A and B ?

Exercise 4 :

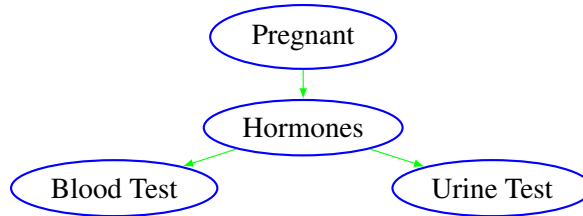
Consider the network defined by the two binary variables A and B , where A is the parent of B . Assume that the conditional probability tables are given as $P(A) = (0.1, 0.9)$ and

	A	
	a_1	a_2
b_1	0.05	0.2
b_2	0.95	0.8

- Assume that you want to estimate $P(b_1)$ using sampling. How many samples would be required if you only accept an error larger than 0.15 in 10% of the cases?
- Generate two random samples, using the following list of random numbers (any time that you use a random number pick one from the list): 0.5, 0.3, 0.01, 0.8.
- Suppose that after sampling 10 states, you got $(A = a_2, B = b_2)$ 3 times, and $(A = a_1, B = b_2)$ 6 times, and $(A = a_1, B = b_1)$ once. Estimate $P(B = b_1)$
- (Optional) Implement the network above in Hugin and use Hugin to sample the number of cases that you calculated in step (i); use the function 'Simulate cases' under 'File'. Use the sampled cases to estimate $P(b_1)$ and compare the result with Hugin. Feel free to use a spreadsheet for the counting.
- Assume that you want to use rejection sampling to estimate $P(A|B = b_1)$. How many samples do you expect you would have to generate in order to end up (after rejection) with a sample set of 1000 cases for estimating the probability.

Exercise 5 :

Consider the insemination example from Section 3.1.13 in BNDG:



Let the probabilities be as in Table 1 ($Ho = y$ means that hormonal changes have taken place) $P(Pr=y) = 0.87$ and $P(Pr=n) = 0.13$.

	$Pr = y$	$Pr = n$		$Ho = y$	$Ho = n$
$Ho = y$	0.9	0.01	$BT = y$	0.7	0.1
$Ho = n$	0.1	0.99	$BT = n$	0.3	0.9

	$Ho = y$	$Ho = n$
$UT = y$	0.8	0.1
$UT = n$	0.2	0.9

Table 1: Tables for Exercise 5.

- What is $P(Pr | BT = n, UT = n)$?
- (Optional) Verify your solution modelling the BN in Hugin.

Exercise 6 :

Complete Exercise 9.4(a-b) in PM.

Exercise 7 :

Consider the running example of the lecture with the following probability tables:

	Burglary			Earthquake	
	t	f		t	f
	.1	.9		.1	.9

		Alarm	
Burglary	Earthquake	t	f
t	t	.9	.1
t	f	.8	.2
f	t	.5	.5
f	f	.1	.9

Alarm	JohnCalls		Alarm	MaryCalls
	t	f	t	f
t	.8	.2	.7	.3
f	.1	.9	.1	.9

Determine the conditional probability $P(MC \mid B = t)$. For each step indicate the operations that are performed.
